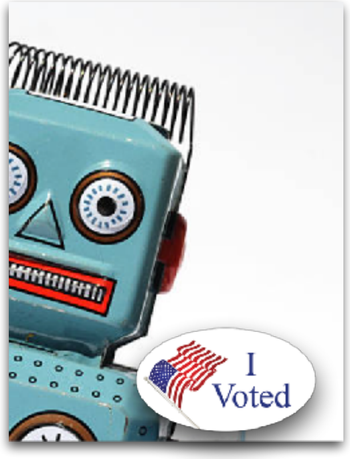
Modeling Democracy

Lecture 14 - **Introduction to redistricting**



Moon Duchin, Winter 2026 — Mon Feb 23

HOW REDISTRICTING **IS** AND **IS NOT** AN OPTIMIZATION PROBLEM

Moon Duchin

Integer Programming and Combinatorial Optimization 2021

adapted from minicourse

Goals for Lectures 14-15

set up the problem

combinatorial explosion

Markov chain approaches

spanning tree ReCom

Future topics, if there's interest

more spanning trees

perfect matchings

IP approaches

ZDDs

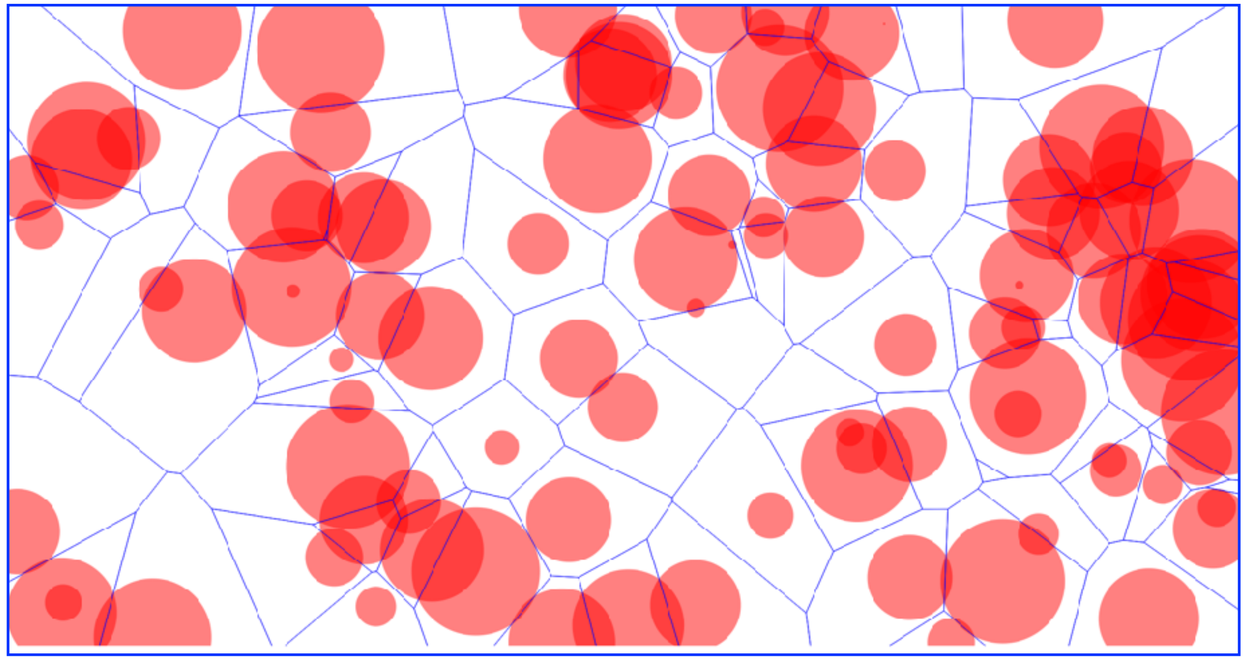
WARMUP

GERRYMANDERING 101

- Every ten years, there's a Census, then we apportion the 435 House seats to the states by population.

- E.g., divide IL into 18 districts subject to
  - equal population
  - contiguity
  - "compactness" (nice shapes)

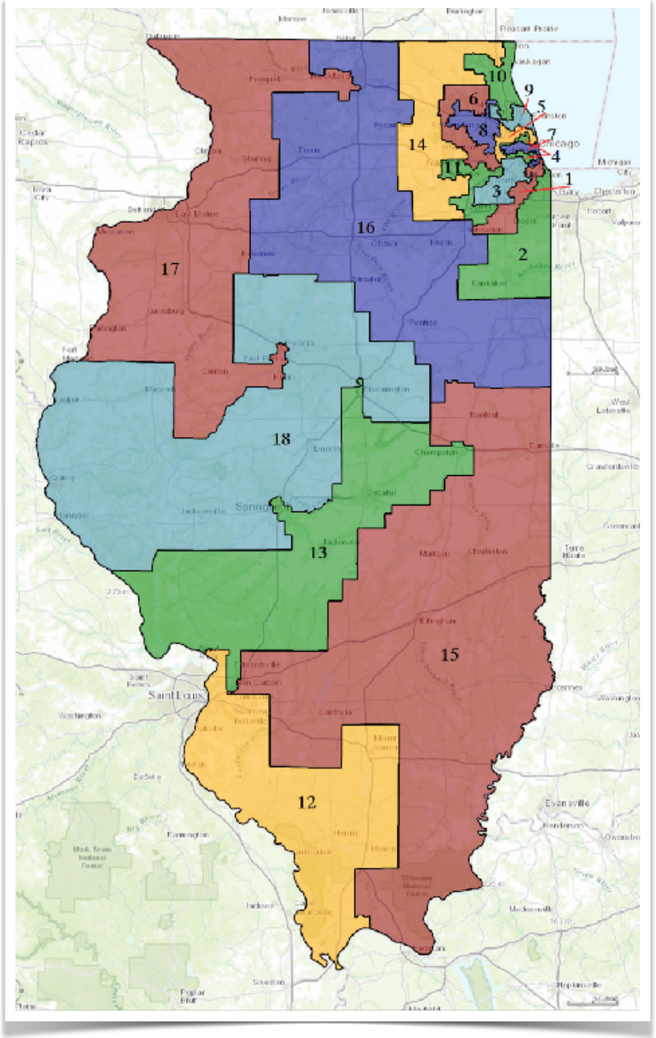
this sounds like something we know how to approach



GERRYMANDERING **201**

- Every ten years, there's a Census, then we apportion the 435 House seats to the states by population.

- E.g., divide IL into 18 districts subject to
    - equal population
    - contiguity
    - "compactness" (nice shapes)
    - preservation of counties, cities, etc
    - respect for "communities of interest"
    - Voting Rights Act of 1965
    - and much more

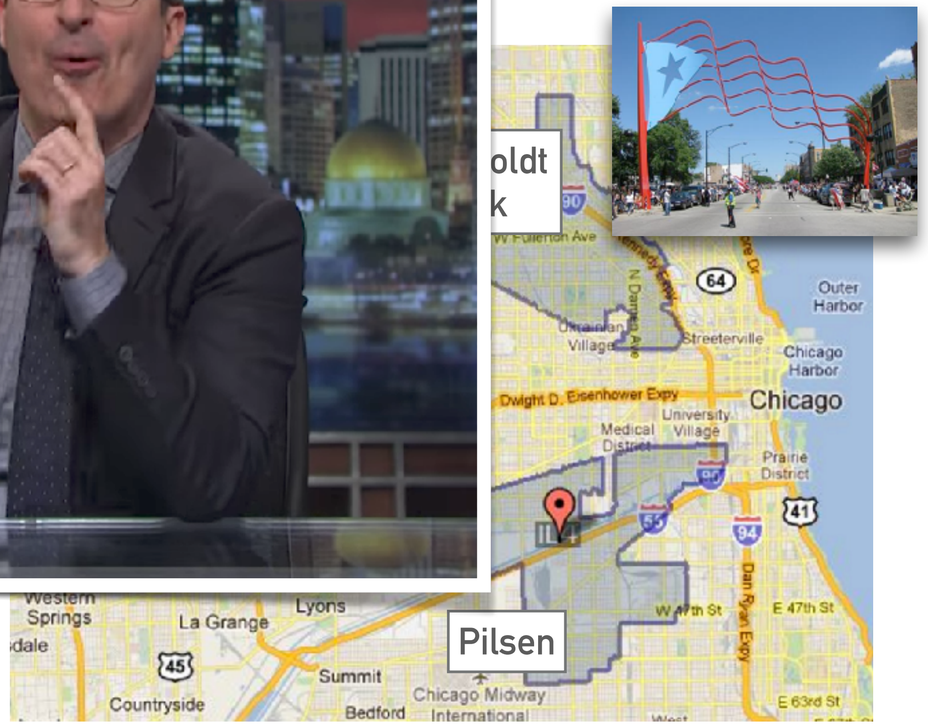


Illinois 4th district

Year	Democrat	Votes	Pct
1992	Luis Gutiérrez	90,452	77.6%
1994	Luis Gutiérrez (inc.)	46,695	75.2%
1996	Luis Gutiérrez (inc.)	85,278	93.6%
1998	Luis Gutiérrez (inc.)	54,244	81.7%
2000	Luis Gutiérrez (inc.)	89,487	88.6%
2002	Luis Gutiérrez (inc.)	67,339	79.7%
2004	Luis Gutiérrez (inc.)	104,761	83.7%
2006	Luis Gutiérrez (inc.)	69,910	85.8%
2008	Luis Gutiérrez (inc.)	112,529	80.6%

Packing!

…but it turns out to be **friendly** packing



DETECTION PROBLEM
VS
PROPOSAL PROBLEM

THE DETECTION PROBLEM

naive flags: bad shapes, disproportion

- bad shapes only flag a *story*
- a requirement for good shapes is not very constraining
- proportionality doesn't come for free; proportionality can be hard or impossible to achieve
- want to create a test that flags abuse but leaves room for human discretion

THE PROPOSAL PROBLEM

best maps? **good** maps? existence proofs?

- redistricting is stubbornly multi-objective
- rules don't come with operational formulas—in law, vagueness can be "a feature, not a bug"
- even if we could build a compound objective function, finding global optimum is likely intractable
- and even if we **had** global optimum, unclear that being better in the 5th decimal place is a reason for choosing one map over other contenders